

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/378528773>

Accurate Vehicles Detection and Speed Estimation Using Homography Based Background Subtraction and Deep Learning Approaches

Conference Paper · December 2023

DOI: 10.1109/ICCIT60459.2023.10441114

CITATIONS

0

READS

233

4 authors, including:



Md. Tohidul Islam

University of Rajshahi

10 PUBLICATIONS 72 CITATIONS

SEE PROFILE



Md. Golam Rashed

University of Rajshahi

66 PUBLICATIONS 399 CITATIONS

SEE PROFILE



Dipankar Das

University of Rajshahi

84 PUBLICATIONS 577 CITATIONS

SEE PROFILE

Accurate Vehicles Detection and Speed Estimation Using Homography Based Background Subtraction and Deep Learning Approaches

Jabbarul Islam, Md. Tohidul Islam, Md. Golam Rashed, and Dipankar Das

Department of Information and Communication Engineering, University of Rajshahi, Bangladesh

email: jabbarulislam7@gmail.com, tanzid1971@gmail.com, golamrashed@ru.ac.bd, dipankar@ru.ac.bd

Abstract—In the context of the ever-evolving landscape of information and communication technology, urban populations worldwide are increasingly embracing the notion of smart cities. Smart transportation, a fundamental component of the smart cities, falls under the purview of what is commonly referred to as the Intelligent Transportation System (ITS). This system plays a pivotal role in the management of highway transportation infrastructure. A key facet of this system involves the widespread installation of CCTV cameras on urban thoroughfares. These cameras serve a dual function, diligently monitoring traffic conditions and detecting anomalies such as traffic congestion and violations of prescribed vehicle speed limits. This research study is primarily dedicated to the task of estimating vehicle speeds using two distinct methodologies: background subtraction based approach and deep learning based approach. Both methodologies leverage the concept of homography based inverse perspective projection to achieve precise vehicle detection and speed estimation. The study is underpinned by the utilization of two distinct datasets, one for the purpose of training and the other for estimating vehicle speeds. The initial phase of our investigation focuses on the accurate detection of vehicles. To accomplish this, we trained a YOLOv8 model, yielding impressive outcomes. In the realm of vehicle detection accuracy, the background subtraction based method achieved an accuracy rate of 92.14%, while the deep learning approach demonstrated an even higher level of accuracy, standing at around 98.88%. Then, our research shifts its focus to the tracking of vehicles frame-by-frame and the subsequent calculation of their speeds. This is achieved with the aid of two reference lines in both vehicle detection methodologies. The results obtained from our experiments unequivocally highlight the superior speed estimation capabilities of deep learning approaches when compared to the background subtraction based method.

Keyword- *Background Subtraction, Deep Learning, Inverse Perspective Mapping, Vehicles Detection, Vehicles Tracking, Speed Estimation*

I. INTRODUCTION

As urban populations continue to expand, daily life faces an increasing array of challenges, particularly in areas where road traffic congestion becomes a significant issue due to high demand and inadequate road capacity and infrastructure. Efficiently addressing these problems is crucial to mitigate their impact effectively. To implement a smart city, intelligent traffic management system is one of the key factors that includes vehicle detection, tracking and speed estimation. Over speed in the speed limited area is the primary cause among various factors contributing to road accidents. Therefore, enforcing speed

limits becomes essential in order to reduce speed-related accidents. However, real-time vehicle speed estimation from video images is a challenging task due to different environmental factors. In a road and highway, certain speed limits condition are imposed in different areas of the road to reduce the road accident, such as turning points, curvature areas, crowded area, predestine crossing area etc. Unfortunately, some drivers tend to disregard traffic speed rules and exceed the speed limit, leading to cause accidents. Traditional methods using low-cost cameras have proven insufficient for real-time detection of such speed violations, as their speed estimation accuracy is generally poor [1], [2]. This inadequacy has motivated the exploration of alternative approaches, combining background subtraction and deep learning methods, to estimate vehicle speed accurately. Additionally, to overcome problem due to the perspective transformation of the camera model which results in varying vehicle aspect ratios, a homography based inverse perspective transformation approach has been proposed and applied in conjunction with the background subtraction and deep learning approaches. The findings indicate that the proposed inverse perspective transformation-based techniques yield comparable results. The proposed system encompasses several noteworthy contributions:

- Introduction of a novel homography based inverse perspective mapping approach.
- Implementation of a method for detecting multiple vehicles.
- Development of a vehicle tracking technique.
- Implementation of a technique for estimating the speed of moving single or multiple vehicles in a video sequence using the tracked vehicles.
- Comparison of different speed estimation methods.
- Creation of a ground truth speed estimation database to facilitate a comparative analysis of our approach.
- Utilization of existing databases to compare and validate our experimental results.

II. RELATED WORKS

Various researchers have explored different approaches to address the challenges of vehicle detection and speed estimation in traffic monitoring systems. In the following subsections, we review the related works of vehicle detection and speed estimation techniques.

A. Vehicle Detection

A video analysis technique for object detection based on the object's position relative to the camera is proposed in [3]. The method involves defining a Region of Interest (ROI) by considering the objects present in the frame and their distance from the camera. Although the algorithm has been enhanced to handle certain weather conditions, it lacks the ability to track vehicles when they change their directions. In [4], Arinaldi et al. presented a traffic video analysis system based on computer vision techniques. They developed two models: one using MoG + SVM and the other based on Faster RCNN, a deep learning architecture. Their findings revealed that Faster RCNN outperformed MoG in detecting static, overlapping, and night-time vehicles. Additionally, Faster RCNN showed superior performance over SVM in classifying vehicle types based on appearances. In 2015, Banu et al. [5] suggested the use of Histogram of Gradient (HoG) feature extraction technique and morphological operations to improve the detection rate. Juan Du's paper [6] provides an overview of object detection techniques based on the CNN family, including CNN, R-CNN, and YOLO algorithms. The paper compares the efficiency of these approaches and highlights the introduction of the YOLO algorithm as a means to enhance efficiency. Recently, Hu et al. [7] proposed a novel algorithm based on YOLOv4 to address speed-related challenges in detecting vehicle targets within video streams. Their approach aimed to improve the efficiency of the detection process.

B. Speed Estimation

To estimate vehicle speed through image processing, Pour Mohammad and Dehghani [8] employed background subtraction and color brightness intensity estimation. However, their system had an accuracy rate of $\pm 10\%$ and only performed data retrieval under normal conditions. Arash Golami R. [9] utilized the CVS method with Euclidean distance to detect vehicle speed in video image sequences, achieving a 90% accuracy rate. Jinhui Lan [10] employed gray constraint optical flow and three frames differencing to measure vehicle speed, obtaining an impressive 99% accuracy. Dogan et al. [11] implemented the Lucas-Kanade optical flow algorithm for vehicle tracking, while Vamsi Krishna et al. [12] used spherical projection and the Lucas Kanade Tomasi method for vehicle detection and speed estimation. However, optical flow algorithms have high complexity and are sensitive to noise. Moreover, these algorithms were limited to handling one vehicle at a time. A method for calculating vehicle speed by transforming 2D images into 3D representations and tracking vehicle positions is presented in [13]. One drawback of their system is that the speed estimation algorithm lacks validation with actual speed, making it difficult to determine accuracy. Alexander A.S. Gunawana et al. [14] proposed a Direct Linear Transformation (DLT) method combined with background subtraction and Mixture of Gaussian (MoG) for vehicle position recognition and speed evaluation. Although their system achieved good accuracy and was not influenced by vehicle shadows, it was unable to handle multi-vehicle detection. Hardy Santosa and

Agus Harjoko [15] and Asif Khan [16] utilized the Euclidean distance for vehicle speed estimation, achieving accuracies of 90.5% and 98.12%, respectively. However, their studies did not investigate the impact of camera angles and ROI selection on speed estimation accuracy. Huang et al. [17] presented YOLOv3, a single-stage deep neural network that exhibited enhanced real-time detection accuracy when tested on diverse datasets. Their research showcased the algorithm's ability to perform well in different environmental conditions.

III. PROPOSED SPEED ESTIMATION TECHNIQUE

This section outlines the key steps of our proposed techniques, which involve vehicle detection and speed estimation. We adopt both background subtraction and deep learning based approaches in our methodology. Initially, we preprocess the real-time video, ensuring the preservation of the vehicle's aspect ratio through homography based inverse perspective transformation. In the background subtraction based approach, we leverage fundamental techniques such as background subtraction, thresholding, and various morphological transformations to detect vehicles. Subsequently, by identifying contours and their centers, we determine the speed of each vehicle.

In contrast, the deep learning approach involves training the YOLOv8 model using a custom dataset specifically designed for vehicle detection. Lastly, we track the bounding box frame-by-frame to accurately estimate the speed of each vehicle. The flowchart is shown in Fig. 1.

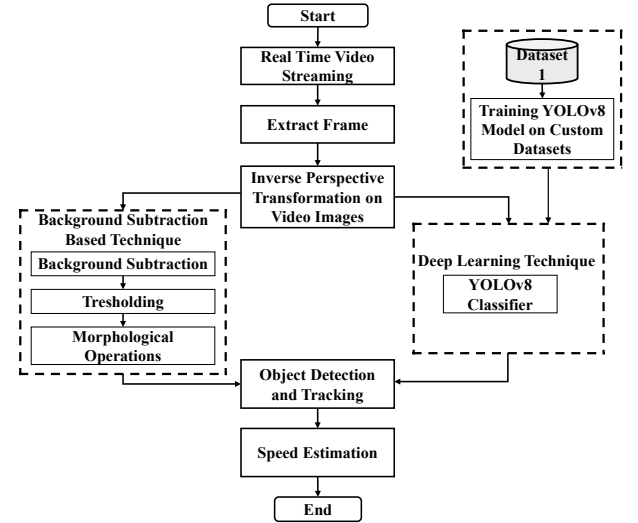


Fig. 1. Vehicle detection and speed estimation algorithm organization.

A. Data Collection and Pre-Processing

In our research, we leveraged two distinct datasets to achieve our objectives: one for training our deep learning model and the other for estimating speed. For the training phase, we acquired five video clips, each lasting 2 minutes, from online sources [18]–[22]. These videos all operated at a consistent frame rate of 30 frames per second, and we

standardized the frame size to 1020x900 pixels using the shape function. Subsequently, we meticulously extracted a total of 3,000 frames from these videos. To facilitate the training process, we annotated these frames using the Alpha Make Sense tools, carefully marking and categorizing objects of interest. We then saved all annotated files in YOLO format, organized into different classes, ensuring that each object was appropriately labeled.

For the purpose of speed estimation, we curated a dataset comprised of authentic video footage captured at different locations from foot-over bridge. This dataset consisted of five video clips, each spanning a duration of around five minutes, all videos are recorded at a consistent frame rate of 30 frames per second. To maintain uniformity, we standardized the frame size to 1020x900 pixels using the shape function. During the meticulous data collection process, we placed a significant emphasis on ensuring the precision of location and speed information. To achieve this, we employed a Google GPS tracker, which provided accurate geospatial data. Furthermore, to enhance the accuracy of our speed estimation procedure, we meticulously synchronized the recorded video footage with our speed measurement application. This synchronization was achieved through the utilization of a clock, ensuring that the speed data was precisely aligned with the corresponding video frames, thus bolstering the overall reliability of our speed estimation methodology.

B. Homography Based Inverse Perspective Mapping

Inverse Perspective Transform ensures accurate image alignment by rectifying distortions in Vehicle Speed Estimation. It adjusts the vehicle's aspect ratio for more precise speed estimation near and far from the Camera. This approach mitigates inaccuracies in frame-by-frame tracking due to changing aspect ratios, enhancing speed estimation accuracy. The basic equation of inverse perspective transformation is:

$$p' = Mp \quad (1)$$

Here, p' defines the transformed points, M defines the Homography matrix and p defines the input points. Initially, the elements of the Homography matrix, M are computed using four point correspondence between original and transformed images (Fig. 2(a-b)). Once the Homography matrix is obtained, the remaining points in the source image can be transformed using the derived matrix. The transformed images before perspective projection are depicted in Fig. 2(c), while the generated images after perspective transformation are shown in Fig. 2(d).

C. Vehicles Detection

Upon implementing the inverse perspective transformation to improve vehicle speed estimation, the next crucial step involves detecting and tracking the vehicles frame-by-frame. For this purpose, we have devised two distinct approaches for vehicle detection, which are outlined as follows.

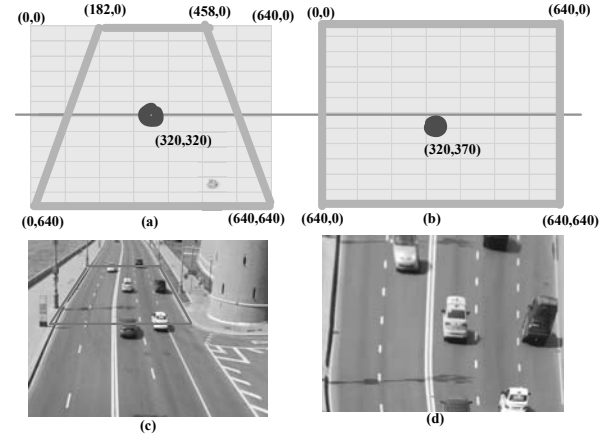


Fig. 2. Perspective transformation (a) Coordinates of the ROI (b) Transformed coordinates (c) Input image with ROI (d) Transformed image.

1) *Background Subtraction Based Approach:* In the background subtraction based technique for vehicle detection, the image is initially transformed into a grayscale image to eliminate unnecessary noise. Subsequently, GaussianBlur is employed to diminish noise and sharp transitions. Background subtraction is then applied to extract the foreground, followed by thresholding to enhance the detection process. Lastly, morphological operations are utilized to further refine the results, ultimately leading to accurate vehicle detection.

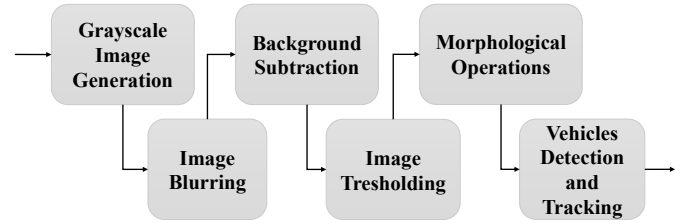


Fig. 3. Vehicle detection pipeline using background subtraction based technique.

2) *Deep Learning Based Approach:* For vehicle detection using deep learning based techniques, the YOLO model is chosen due to its leading performance in various aspects, including speed estimation, detection accuracy, good generalization, and being open-source. Among the different versions of YOLO, YOLOv8 stands out as a preferable choice for this research because of its improved accuracy compared to previous YOLO models. The proposed vehicle detection pipeline using deep learning based technique is shown in Fig. 4. For accurate detection, the YOLOv8 Model is trained on custom dataset.

D. Vehicles Tracking

Vehicle tracking holds a significant role in precise speed estimation as it enables the calculation of actual vehicle speeds. In this research, vehicle tracking is conducted on a frame-by-frame basis by utilizing the center point of each vehicle's bounding box. The distance between consecutive

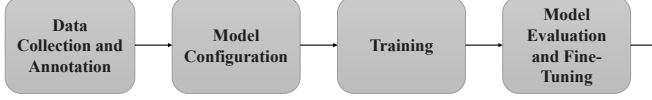


Fig. 4. Vehicle detection pipeline using deep learning based technique.

center points is then computed using the Euclidean formula. By applying a threshold value, the tracking process accurately identifies the vehicles and assigns unique IDs. Fig. 5 provides an illustration of the distance calculation and the tracking process. The Euclidean formula to measure distance between two points is given by the Eq. 2.

$$pd = \sqrt{(m2 - m1)^2 + (n2 - n1)^2} \quad (2)$$

Here, $(m1, n1)$ and $(m2, n2)$ denoted as the center point coordinates of the first frame and second frame of the same vehicles and pd defines their pixels distance.

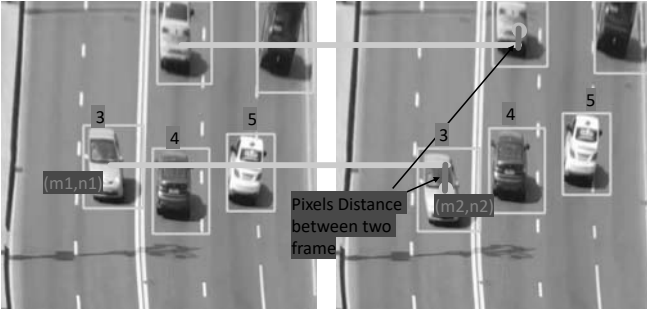


Fig. 5. Vehicle tracking in two subsequent images.

E. Speed Estimation

The ultimate phase involves assessing the velocity of each moving object within the video stream. This entails precisely gauging the displacement between two specific reference points along the captured path in the video feed. Upon a car's passage through the initial reference point, a dedicated timer is initiated for that particular vehicle. Likewise, when the vehicle traverses the second reference point, the timer is halted, and the time differential is computed, representing the duration the vehicle required to travel from the initial to the second reference point.

$$d = \frac{r * (l2 - l1)}{h} \quad (3)$$

Here, d is the distance in meter, r is the real-world distance in meter, h is the image height(pixels) and $(l2 - l1)$ represents pixel difference between two lines. Finally, Eq. 4 is used to calculate the speed(km/h) of the car.

$$v = \frac{d}{t} * 3.6(km/h) \quad (4)$$

Here, t represents the total moving time between two lines and v is the final speed of the vehicle.

IV. RESULTS AND DISCUSSION

To assess the effectiveness of our proposed method, we've gathered real-world data from the environment. In the subsequent sections, we delve into the processes of acquiring data, detecting and tracking vehicles, and ultimately estimating their speeds.

A. Training YOLOv8 Model with Custom Dataset

In our deep learning approach, we used the dataset described in Section 2. We combined annotated images and text files to create a comprehensive dataset, which we stored in a cloud drive for easy access and collaboration. We split the dataset into 80% for training our model and 20% for validation, ensuring effective performance evaluation. This meticulous partitioning was crucial for successful model training. The training process utilized both CPU and GPU resources, with Google Colab providing a free GPU environment for efficient training. Dependencies and prerequisites were installed beforehand to ensure a smooth execution of the model. The Fig. 6 shows the mAP and loss plots after training the YOLOv8 small model on the vehicles detection dataset.

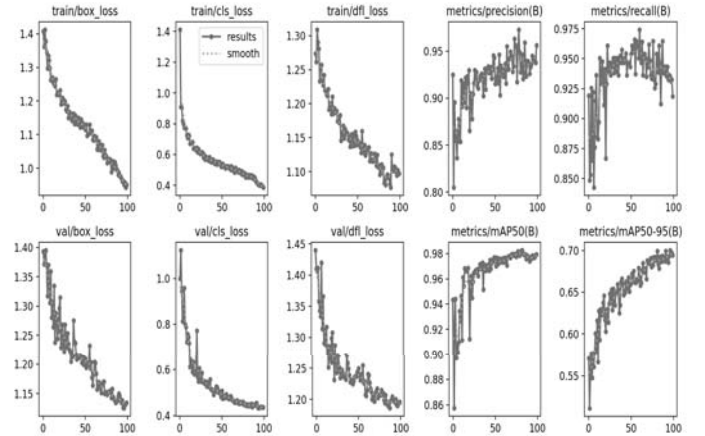


Fig. 6. mAP and loss plots after training the YOLOv8 small model on the vehicles detection dataset.

The trained model undergoes a meticulous evaluation process on an independent test dataset to gauge its proficiency in accurately detecting vehicles. This evaluation aims to measure the model's precision and efficacy in identifying and quantifying vehicles within the provided dataset.

B. Detection and Tracking

As mentioned earlier, both methods begin with the initial step of employing inverse perspective projection to detect vehicle accurately and calculate speed of the detected vehicle.

In the background subtraction based method as shown in Fig. 7, we start by converting the image to grayscale to reduce processing time. We then apply GaussianBlur to smooth out sudden variations. Background subtraction is used to extract

the foreground, and thresholding is used to enhance the detection accuracy. Finally, morphological operations refine the results, ensuring accurate vehicle detection.

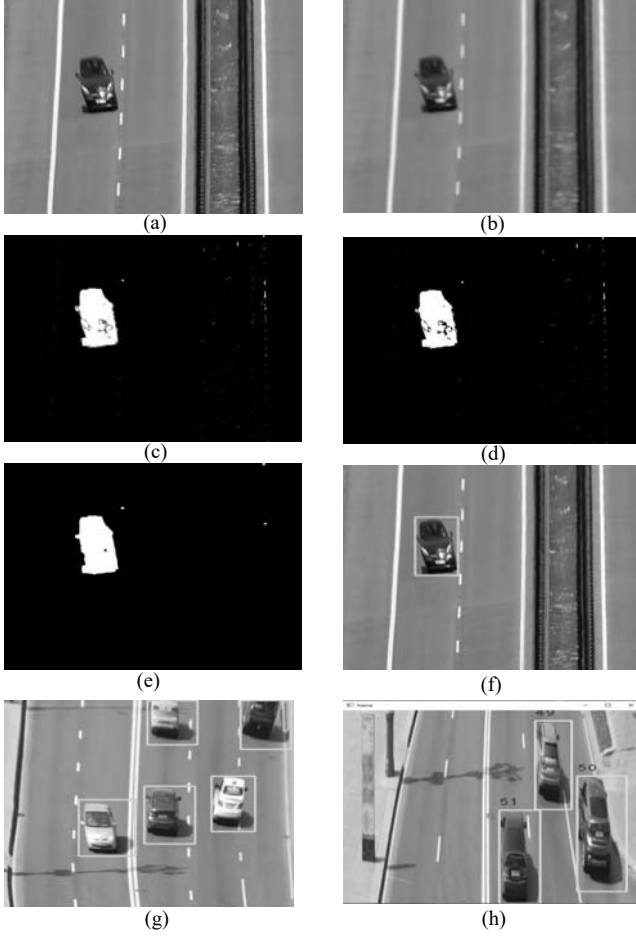


Fig. 7. Vehicle detection (a) Gray image (b) Blur image (c) Background subtraction (d) Image thresholding (e) Morphological operations (f-h) Final detection result.

We have successfully achieved nearly perfect vehicle detection using the trained model depicted in Fig. 8 through our deep learning based approach. We also conducted an experiment involving vehicle detection in five videos using two distinct techniques, as demonstrated in Table I. In the background subtraction based technique, we observed that when two vehicles moved very close to each other, they appeared as a single detection due to the shadow effect. As a result, the detection accuracy (F-1 score) was 93.10%. On the other hand, the deep learning-based approach didn't encounter this issue, as it relied on a trained model for detection. Consequently, the detection accuracy was significantly higher at approximately 98.33%.

C. Speed Measurement

In this subsection, we have explored the estimation of vehicle speed through the application of two proposed methods. To validate the accuracy of our approaches, we have employed

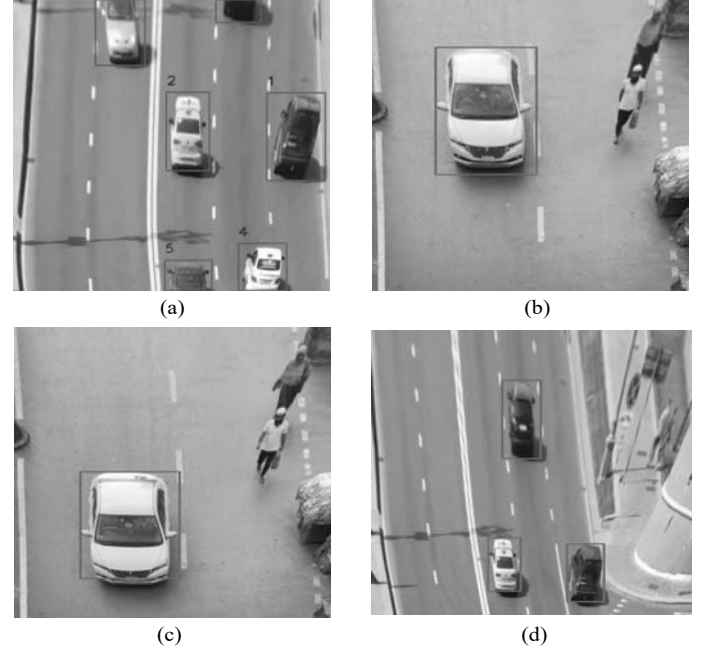


Fig. 8. Vehicle detection using trained YOLOv8 model.

TABLE I
COMPARISON OF DETECTED VEHICLES VS. ACTUAL VEHICLES USING BACKGROUND SUBTRACTION AND DEEP LEARNING BASED APPROACHES

Video#	Positive Sample	True Positive		False Positive	
		Meth. 1	Meth. 2	Meth. 1	Meth. 2
Video-1	30	27	30	2	1
Video-2	15	15	15	0	1
Video-3	23	20	22	1	0
Video-4	7	7	7	0	0
Video-5	14	12	14	1	0
Total	89	81	88	4	2

ground truth values for comparison. Our experimentation encompassed two different videos, allowing us to determine the most effective method. The experimental outcomes, as presented in Table II and Table III, showcase the actual and estimated speeds of the vehicles in these videos. Method 1 represents background subtraction based techniques, while Method 2 signifies deep learning approaches. The results reveal that both methods exhibit remarkably low error rates, with deep learning methods demonstrating notably superior performance when compared to image processing techniques.

TABLE II
EXPERIMENTAL RESULTS FOR SPEED ESTIMATION ON VIDEO 1

Car ID.	Actual Speed (km/h)	Method 1		Method 2	
		Measured Speed (km/h)	Diff.	Measured Speed (km/h)	Diff.
1	26.4	25.5	0.9	25.8	0.6
2	16.0	16.2	0.2	16.4	0.4
3	32.5	34.0	1.5	31.0	1.5
4	34.0	35.2	1.2	34.2	0.2
5	44.0	43.5	0.5	43.0	1.0

TABLE III
EXPERIMENTAL RESULTS FOR SPEED ESTIMATION ON VIDEO 2

Car ID.	Actual Speed (km/h)	Method 1		Method 2	
		Measured Speed (km/h)	Diff.	Measured Speed (km/h)	Diff.
1	20.5	19.5	1.0	22	1.0
2	10.0	9.0	1.0	9.5	0.5
3	36.5	34.0	2.5	37.0	0.5
4	47.0	45.5	1.5	47.0	0.0
5	41.0	42.0	1.0	43.0	2.0

To assess the performance of our speed estimation method, we conducted a comparative analysis by evaluating error rates for four different methods as shown in Table IV. Our evaluation demonstrates that our system performs exceptionally well, exhibiting lower error rates when compared to these four methods. Notably, among our two proposed methods, the deep learning approach consistently delivers the best performance.

TABLE IV
ERROR RATES FOR DIFFERENT SPEED ESTIMATION METHODS

Speed Range (km/h)	Error Rate in Other Methods (%)				Error Rate in Proposed Method (%)	
	[23]	[24]	[1]	[2]	Method 1	Method 2
10-20	8.33	6.29	-	-	7.19	4.70
20-30	7.47	5.78	-	-	7.55	4.6
30-40	6.0	-	8.7	-	3.55	4.78
40-50	4.35	-	-	6.0	2.81	2.93
50-60	-	-	-	7.75	-	-
Average	6.54	6.04	8.70	6.88	5.28	4.25

V. CONCLUSIONS

Based on the work conducted thus far, it can be concluded that speed estimation using computer vision holds significant potential in real-world applications. The proposed algorithm has demonstrated a good accuracy in estimating speed of the vehicle. This accuracy is primarily influenced by the characteristics of detected blobs and their consistency across consecutive frames. Additionally, the stability of camera frame rates plays a crucial role in optimizing the algorithm's performance.

Utilizing Foreground Shape Analysis (FSA) in intensity-level images offers distinct advantages over traditional foreground extraction techniques. It reduces the need for excessive adjustments to the extracted foreground, preventing the estimated speed from becoming excessively erratic due to centroid position fluctuations caused by irregularities in blob shapes in subsequent frames. The selection of FSA, owing to its ability to maintain similarity in blob shapes across consecutive frames, further enhances the stability of the running mean, contributing to improved accuracy in speed estimation.

REFERENCES

[1] N. Raikar, "Development of vehicle speed estimation technique using image processing," *International Journal of Engineering Research and*, vol. V9, 09 2020.

[2] C.-J. Lin, J. Shiou-Yun, and H.-W. Lioa, "A real-time vehicle counting, speed estimation, and classification system based on virtual detection zone and yolo," *Mathematical Problems in Engineering*, vol. 2021, pp. 1–10, 11 2021.

[3] P.-S. Liao, T.-S. Chen, and P. C. Chung, "A fast algorithm for multilevel thresholding," *J. Inf. Sci. Eng.*, vol. 17, pp. 713–727, 2001.

[4] A. Arinaldi, J. A. Pradana, and A. A. Gurusinga, "Detection and classification of vehicles for traffic video analytics," in *INNS Conference on Big Data*, 2018.

[5] A. S. Banu and P. Vasuki, "Video based vehicle detection using morphological operation and hog feature extraction," *ARNP Journal of Engineering and Applied Sciences*, vol. 10, no. 5, pp. 1866–1871, 2015.

[6] J. Du, "Understanding of object detection based on cnn family and yolo," *Journal of Physics: Conference Series*, vol. 1004, no. 1, p. 012029, apr 2018. [Online]. Available: <https://dx.doi.org/10.1088/1742-6596/1004/1/012029>

[7] X. Hu, Z. Wei, and W. Zhou, "A video streaming vehicle detection algorithm based on yolov4," in *2021 IEEE 5th Advanced Information Technology, Electronic and Automation Control Conference (IAEAC)*, vol. 5, 2021, pp. 2081–2086.

[8] A. Dehghani and A. Pourmohammad, "Single camera vehicles speed measurement," *2013 8th Iranian Conference on Machine Vision and Image Processing (MVIP)*, pp. 190–193, 2013.

[9] A. G. Rad, A. Dehghani, and M. R. Karim, "Vehicle speed detection in video image sequences using cvs method," *International Journal of Physical Sciences*, vol. 5, pp. 2555–2563, 12 2010.

[10] J. Lan, J. Li, G. Hu, B. Ran, and L. Wang, "Vehicle speed measurement based on gray constraint optical flow algorithm," *Optik*, vol. 125, no. 1, pp. 289–295, 2014. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0030402613009005>

[11] S. Dogan, M. S. Temiz, and S. Külür, "Real time speed estimation of moving vehicles from side view images from an uncalibrated video camera," *Sensors (Basel, Switzerland)*, vol. 10, pp. 4805 – 4824, 2010.

[12] V. K. Madasu and M. Hanmandlu, "Estimation of vehicle speed by motion tracking on image sequences," *2010 IEEE Intelligent Vehicles Symposium*, pp. 185–190, 2010.

[13] H. A. Rahim, R. B. Ahmad, A. S. A. M. Zain, and U. U. Sheikh, "An adapted point based tracking for vehicle speed estimation in linear spacing," *International Conference on Computer and Communication Engineering (ICCCE'10)*, pp. 1–4, 2010.

[14] A. A. S. Gunawan, D. A. Tanjung, and F. E. Gunawan, "Detection of vehicle position and speed using camera calibration and image projection methods," *Procedia Computer Science*, vol. 157, pp. 255–265, 2019.

[15] H. Sundoro and A. Harjoko, "Vehicle counting and vehicle speed measurement based on video processing," *Journal of Theoretical and Applied Information Technology*, vol. 84, no. 2, pp. 233–241, 02 2016.

[16] A. Khan, D. S. Z. Sarker, and S. Rayamajhi, "Speed estimation of vehicle in intelligent traffic surveillance system using video image processing," *International journal of scientific and engineering research*, vol. 5, pp. 1384–1390, 2014.

[17] Y.-Q. Huang, J.-C. Zheng, S.-D. Sun, C.-F. Yang, and J. Liu, "Optimized yolov3 algorithm and its application in traffic flow detections," *Applied Sciences*, vol. 10, no. 9, p. 3079, 2020.

[18] K. Majek, "4k road traffic video for object detection and tracking-free download now!" *YouTube*, July 2018. [Online]. Available: <https://tinyurl.com/5dtmd8dv>

[19] A. Nikishaev, "Road traffic video for object recognition," *YouTube*, Sept. 2017. [Online]. Available: <https://tinyurl.com/b7ea6s35>

[20] STRAMS, "4k road traffic video for object detection and tracking," *YouTube*, June 2021. [Online]. Available: <https://tinyurl.com/4upzvmme>

[21] N. Martinez, "4k video of highway traffic!" *YouTube*, Oct. 2017. [Online]. Available: <https://tinyurl.com/4zprzvbe>

[22] V. K., "Relaxing highway traffic," *YouTube*, July 2017. [Online]. Available: <https://tinyurl.com/bdec33x>

[23] A. Khan, I. Ansari, M. Sarker, and S. Rayamajhi, "Speed estimation of vehicle in intelligent traffic surveillance system using video image processing," *International Journal of Scientific and Engineering Research*, vol. 5, pp. 1384–1390, 12 2014.

[24] M. G. Moazzam, M. R. Haque, and M. S. Uddin, "Image-based vehicle speed estimation," *Journal of Computer and Communications*, vol. 7, pp. 1–5, 2019. [Online]. Available: <https://api.semanticscholar.org/CorpusID:191168627>